ANSWER KEY & MARKING SCHEME SET 1

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** either a magnet or magnetic material	Attraction alone does not prove a magnet.
2	**(a)** iron, nickel, cobalt	Standard magnetic materials.
3	**(d)** push away from each other	Like poles repel.
4	**(d)** north pole	Like poles repel.
5	**(d)** aluminium	Aluminium is not attracted.
6	**(d)** unlike poles	Unlike attract.
7	**(b)** unlike	Opposite poles attract.
8	**(d)** only some metals are magnetic	Only some are magnetic.
9	**(d)** in a box with a keeper	Keepers prevent loss.
10	**(b)** north-seeking pole	North pole.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	No, a magnet repels one pole (1); The rod is a magnetic material only (1).	1+1
12	They repel (1); Like poles repel (1).	1+1
13	Pole (1); Magnetic force is concentrated there (1).	1+1
14	Bring a known magnet to both ends (1); If one end is repelled, it is a magnet (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Use a magnet to remove iron (1); Add water to dissolve salt, filter sand (1); Evaporate water to get salt (1).	1+1+1
16	Bring the marked N pole near an end (1); If attracted, that end is S (1); If repelled, that end is N (1).	1+1+1
17	Aluminium: foils (1); Copper: wires (1); Gold: jewellery (1).	1+1+1
	OR: Magnetise the needle (1); Float it on a cork in water (1); It settles in north-south (1).	1+1+1
18	It is pushed by Earth's field (1); Overshoots north-south (1); Friction stops it finally (1).	1+1+1
	OR: Iron body may deflect it (1); Reading may be wrong (1); Use away from large iron (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Hang one bar magnet (1); Bring N of other near N (1); Observe repulsion (1); Bring S near N: attraction (1); Conclude: like repel, unlike attract (1).	1+1+1+1+1
	OR: Both attracted by the bar (1); Magnet may be repelled (1); Iron is not repelled (1); Test confirms magnet (1); Iron may become magnetised (1).	1+1+1+1+1
20	(a) Force passes through paper (1); (b) Copper is non-magnetic (1); (c) Pairs with keepers preserve magnetism (1); Pole arrangement unlike (1); Prevents weakening (1).	1+1+1+1+1
	OR: Spread paper clips on a table (1); Bring the north pole, the middle and the south pole near them in turn (1); Count the clips lifted each time (1); The ends lift the most clips (1); The middle lifts the fewest (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET 2

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** a magnet can attract iron without touching it	Force acts from a distance.
2	**(a)** compass	Compass.
3	**(a)** south pole	Unlike poles attract.
4	**(b)** magnetic force passes through glass	Non-magnetic materials do not block it.
5	**(a)** south	South end points south.
6	**(c)** the middle	Field lines.
7	**(a)** it weakens	Dropping disturbs its alignment.
8	**(b)** is longer and still has two poles	Two end poles remain.
9	**(c)** the poles	Field is strongest there.
10	**(c)** near the poles	Poles are strongest.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	The magnet makes it a weak magnet (1); Contact with the magnet gives it magnetism (1).	1+1
12	Rust is iron oxide (1); It is still magnetic (1).	1+1
13	Opposite poles attract (1); Force is large when close (1).	1+1
14	Magnetic force can erase data (1); It disturbs circuits (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Place an iron pin in a glass of water (1); Hold the magnet outside the glass (1); The pin moves to the magnet (1).	1+1+1
16	They lift iron scrap (1); Non-magnetic scrap stays (1); They separate scrap easily (1).	1+1+1
17	Use steel (1); Magnetise properly (1); Store with keepers (1).	1+1+1
	OR: Bring each near a magnet (1); The magnet can repel (1); Iron is only attracted (1).	1+1+1
18	Magnetic force acts at a distance (1); Contact force needs touch (1); Example: magnet and nail (1).	1+1+1
	OR: Natural: lodestone (1); Artificial: made from steel (1); Artificial can be shaped (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Needle, magnet, cork, water (1); Magnetise the needle by stroking (1); Float on the cork (1); It rests north-south (1); Compare with a real compass (1).	1+1+1+1+1
	OR: Two magnets (1); Each has N and S (1); Filings gather at ends of each piece (1); No middle clustering (1); Strength is less (1).	1+1+1+1+1
20	Magnet's small parts align (1); Heating disturbs alignment (1); Hammering disturbs it (1); Dropping too (1); Use keepers and gentle handling (1).	1+1+1+1+1
	OR: Magnetic strip stores data (1); Magnets can erase it (1); Keep away (1); Store magnets separately (1); Do not carry in pockets (1).	1+1+1+1+1

Mistakes to avoid:

- Thinking everything metallic is magnetic; copper and aluminium are metals but **not** magnetic.
- Using attraction alone as the test for a magnet (iron is attracted too); the sure test is **repulsion**.
- Writing 'like poles attract'; **like poles repel** and unlike poles attract.
- Thinking the centre of a bar magnet is the strongest part; it is strongest at the **poles**.
- Believing a magnet cut in half gives a separate north and south pole; each piece has both.
- Forgetting that a freely suspended magnet comes to rest in the north-south direction.

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MASTERMIND

ANSWER KEY & MARKING SCHEME SET 3

Chapter 4: **Exploring Magnets** | Class 6 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** the force is concentrated at the poles	Magnetism is strongest at poles.
2	**(b)** stroke it with a magnet in one direction	Stroking aligns small magnets.
3	**(d)** steel disturbs the field	Steel is magnetic.
4	**(c)** stay in any direction	It aligns only due to the field.
5	**(d)** the magnet's field acts on it	Magnetic force.
6	**(a)** magnetic separation	A magnet picks them.
7	**(c)** brass	Brass is an alloy of copper and zinc.
8	**(c)** it holds magnetism for long	Steel retains magnetism.
9	**(d)** iron nail and steel pin	Both are magnetic.
10	**(d)** steel or alloy	Steel alloys give strong magnets.

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Aluminium is non-magnetic (1); Magnets act only on a few materials (1).	1+1
12	It turns to north-south (1); We can mark directions (1).	1+1
13	It closes the poles (1); Prevents loss of magnetism (1).	1+1
14	Heat it (1); Hammer it (1).	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Curves from one pole to the other (1); Dense near the poles (1); Sparse far away (1).	1+1+1
16	Speaker has a magnet (1); Moves a coil (1); Produces sound (1).	1+1+1
17	Pole pointing north (1); Free magnet (1); Called north pole (1).	1+1+1
	OR: Heat disturbs alignment (1); Tiny magnets become random (1); So the magnet weakens (1).	1+1+1
18	It aligns with Earth's field (1); North-south direction known (1); Other directions found (1).	1+1+1
	OR: Magnetic strip in the door (1); Attracts steel body (1); Sealed shut (1).	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	Waste moves on a belt (1); Magnet above lifts steel (1); Non-magnetic items move on (1); Aluminium separated separately (1); Recovery of iron (1).	1+1+1+1+1
	OR: Hold flat (1); Needle settles (1); Rotate dial (1); Find the north (1); Plan the path (1).	1+1+1+1+1
20	Lodestone pointed north-south (1); Sailors used it (1); Made into needle (1); Compass invented (1); Used in navigation (1).	1+1+1+1+1
	OR: Test each with iron filings (1); Wood does not attract (1); Iron attracts but is not repelled (1); Magnet repels one end (1); Draw conclusion (1).	1+1+1+1+1

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